StaffSync HRMS Phase-Wise Implementation Documentation

Executive Overview
StaffSync is an enterprise-grade Human Resource Management System (HRMS) built on a modern full-stack TypeScript architecture (React 19, Tailwind CSS v4, Vite, Express, and Mongoose/Dual-Engine storage). This document details each phase of the project’s inception, design, engineering, integration, and verification.

Table of Phases

Phase	Phase Name	Primary Objectives & Deliverables	Status
Phase 1	Architecture & Data Modeling	Domain entities, TypeScript schemas, API contracts	Completed
Phase 2	Dual Storage Engine	MongoDB Atlas cloud driver + resilient in-memory fallback	Completed
Phase 3	Authentication & RBAC	JWT auth, bcrypt password hashing, 3 role personas	Completed
Phase 4	Attendance & Timesheets	Punch clock, shift tracking, attendance audits, manual corrections	Completed
Phase 5	Multi-Tier Leave Workflow	Leave balances, manager pre-approval, HR final sign-off, cancellation	Completed
Phase 6	Workforce Directory & Ops	Department units, employee profiles, training modules, announcements	Completed
Phase 7	Audit Trail & Analytics	Immutable compliance logging, CSV reports, department metrics	Completed
Phase 8	Theme Engine & Dark Mode	CSS variable mapping, dark mode toggle, zero-contrast defect polish	Completed
Phase 9	Quality Assurance & Testing	Automated 17-point test suite, edge-case validation, health checks	Completed

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Phase 1: Architecture & Data Modeling

1.1 Objectives
Define the core business entities for modern human resource operations, ensuring strict separation of concerns, strong TypeScript typings, and normalized relationship models.

1.2 Core Data Schemas

1. **User Account (‘User‘)**:
   - ‘_id‘, ‘email‘, ‘passwordHash‘, ‘role‘ (‘’hr’ | ’manager’ | ’employee’‘), ‘employeeId‘, ‘isActive‘, ‘createdAt‘, ‘updatedAt‘.
2. **Employee Profile (‘Employee‘)**:
   - ‘_id‘, ‘employeeCode‘ (e.g., ‘EMP-1001‘), ‘firstName‘, ‘lastName‘, ‘email‘, ‘departmentId‘, ‘designation‘, ‘jobTitle‘, ‘role‘, ‘managerId‘, ‘joiningDate‘, ‘employmentType‘ (‘Full-Time‘, ‘Contract‘, ‘Intern‘), ‘salary‘, ‘phone‘, ‘address‘, ‘employmentStatus‘.
3. **Department (‘Department‘)**:
   - ‘_id‘, ‘name‘, ‘code‘, ‘headEmployeeId‘, ‘budget‘, ‘description‘, ‘location‘.
4. **Attendance Record (‘Attendance‘)**:
   - ‘_id‘, ‘employeeId‘, ‘date‘, ‘checkIn‘, ‘checkOut‘, ‘workedMinutes‘, ‘status‘ (‘Present‘, ‘Late‘, ‘Half Day‘, ‘Absent‘, ‘On Leave‘), ‘remarks‘, ‘correctionReason‘, ‘isCorrected‘, ‘correctedBy‘.
5. **Leave Request (‘Leave‘)**:
   - ‘_id‘, ‘employeeId‘, ‘leaveTypeId‘ (‘casual‘, ‘sick‘, ‘annual‘, ‘unpaid‘), ‘startDate‘, ‘endDate‘, ‘totalDays‘, ‘reason‘, ‘status‘ (‘Pending Manager Approval‘, ‘Pending HR Approval‘, ‘Approved‘, ‘Rejected‘, ‘Cancelled‘), ‘approvalHistory‘ array with manager/HR review timestamps, comments, and decision audit.
6. **Leave Balance (‘LeaveBalance‘)**:
   - ‘_id‘, ‘employeeId‘, ‘leaveTypeId‘, ‘allocatedDays‘, ‘usedDays‘, ‘pendingDays‘, ‘year‘.
7. **Training Workshop (‘Training‘)**:
   - ‘_id‘, ‘title‘, ‘description‘, ‘departmentId‘, ‘instructor‘, ‘startDate‘, ‘endDate‘, ‘capacity‘, ‘participants‘ array with progress tracking, completion status, and scores.
8. **Announcement (‘Announcement‘)**:
   - ‘_id‘, ‘title‘, ‘content‘, ‘priority‘ (‘Low‘, ‘Medium‘, ‘High‘, ‘Urgent‘), ‘authorId‘, ‘targetDepartmentId‘, ‘expiresAt‘, ‘isActive‘.
9. **Audit Trail Log (‘AuditLog‘)**:
   - ‘_id‘, ‘action‘, ‘performedBy‘, ‘userRole‘, ‘entityType‘, ‘entityId‘, ‘details‘, ‘ipAddress‘, ‘timestamp‘.

Phase 2: Dual Storage Database Architecture

2.1 Objectives

Eliminate local runtime blocking due to missing cloud credentials or network IP white listing restrictions while providing seamless production MongoDB Atlas persistence.

2.2 Implementation
- **Cloud Driver Connection**:
  - Connects to remote MongoDB Atlas clusters using ‘mongoose.connect(process.env.MONGODB_URI)‘.
  - Configures a 3-second non-blocking timeout with automatic reconnection handling.
- **In-Memory Seeded Fallback**:
  - Automatically activates if Atlas is unreachable or IP is not whitelisted.
  - Seeds pre-populated organizations, departments, roles, active staff, leave histories, attendance timestamps, and audit events.
  - Zero disruption to frontend interfaces; queries and mutations reflect in runtime state.
- **Health Diagnostic Endpoint (‘/api/v1/health‘)**:
  - Exposes realtime status: ‘activeEngine‘ (‘’mongodb’‘ vs ‘’in-memory’‘), collection counts, server uptime, and timestamp.

Phase 3: Authentication & Role-Based Access Control (RBAC)

3.1 Objectives
Deliver enterprise-grade JWT-based authentication paired with role-based routing and permissions for three distinct organizational personas.

3.2 Role Hierarchy
1. **HR Administrator (‘hr‘)**:
   - Full organizational oversight.
   - Employee onboarding, salary viewing, department budgets, final leave approval, company announcements, audit trails, and data exports.
2. **Team Manager (‘manager‘)**:
   - Supervisory control over assigned direct reports.
   - Stage 1 leave approvals, team attendance monitoring, team training oversight.
3. **Employee (‘employee‘)**:
   - Self-service portal: Punch-in/out, personal timesheets, leave balance viewing, leave application, assigned training completion, profile management.

3.3 Security Implementation
- Passwords hashed using ‘bcryptjs‘ with salt rounds.
- Signed JWT tokens (‘staffsync_token‘) stored securely in client state and passed via HTTP ‘Authorization: Bearer <token>‘ headers.
- Route protection middleware (‘authenticateToken‘, ‘requireRole([’hr’, ’manager’])‘).

Phase 4: Attendance & Real-Time Punch-Clock Engine

4.1 Objectives
Provide an accurate, tamper-resistant system for employee daily attendance, shift dur
  ation calculation, and supervisor oversight.

4.2 Features Delivered
- **One-Click Check-In / Check-Out**:
  - Prevents duplicate punches for the same calendar date.
  - Automatically calculates total minutes worked upon check-out.
- **Dynamic Shift Status**:
  - Flags punches after 9:30 AM as ‘Late‘.
  - Flags shifts under 240 minutes as ‘Half Day‘.
- **Manager/HR Timesheet Auditing**:
  - View organization-wide timesheets by date range and department.
  - Manual correction feature with compulsory audit reason logging and supervisor sig
  nature.

Phase 5: Multi-Tier Leave Request & Lifecycle Workflow

5.1 Objectives
Prevent leave conflicts and policy violations through automated balance checks and st
  ructured two-tier manager/HR approval pipelines.

5.2 Workflow Pipeline
’’’
[Employee Submits Request]

[Check Balance: allocated - (used + pending)]

[Status: Pending Manager Approval]

[Manager Rejects]   [Manager Approves]
(Balance refunded)

```
                        [Status: Pending HR Approval]



                     [HR Rejects]         [HR Approves]
                  (Balance refunded)      (Pending -> Used)
```

5.3 Cancellation Lifecycle
- Employees can cancel unapproved pending leaves at any time, immediately releasing t
  he allocated days back to their available pool.

Phase 6: Workforce Directory & Operational Modules

6.1 Objectives
Centralize employee records, organizational divisions, skill building, and corporate
  communications.

6.2 Modules
- **Employee Directory**:
  - Searchable by name, email, department, role, or status.
  - Modal view for personal contact info, joining date, reporting manager, and alloca
  ted compensation.
- **Department Units**:
  - Engineering, Marketing, Human Resources, Product & Design, Sales.
  - Head of department linking and allocated operating budgets.
- **Training Workshops**:
  - Course enrollment, progress percentage tracking, instructor tracking, and pass/fa
  il certificates.
- **Corporate Announcements**:
  - Priority badging (‘High‘, ‘Urgent‘, ‘General‘), target department filtering, and
  expiry dates.

Phase 7: Compliance, Audit Trails & CSV Reporting

7.1 Objectives
Ensure regulatory compliance (SOX, GDPR internal controls) and operational visibility

through immutable logs and one-click data exports.

7.2 Features
- **Audit Trail Logging**:
  - Automatically records every security-sensitive mutation (logins, leave reviews, attendance adjustments, profile updates).
  - Displays user ID, role, action, target entity, timestamp, and metadata diffs.
- **Authorized CSV Exports**:
  - One-click downloads for Workforce Roster, Attendance Records, Leave Utilization, and Audit Event Logs.

Phase 8: Theme Engine & Dark Mode

8.1 Objectives
Provide an accessible, eye-friendly dark mode experience that adheres to modern WCAG contrast ratios across all dashboards, tables, dialogs, and navigation controls.

8.2 Deliverables
- **Tailwind CSS v4 Integration**:
  - Configured `@custom-variant dark (&:where(.dark, .dark *));` in `src/index.css`.
- **State Management**:
  - Created `ThemeContext` persisting state to `localStorage.staffsync_theme` and toggling `.dark` on `document.documentElement`.
- **Universal Semantic Styling**:
  - Mapped slate background and border utilities (`bg-white` -> `bg-slate-900`, `bg-slate-50` -> `bg-slate-950`).
  - Added dedicated theme toggle controls to both the **Top Navigation Bar** and **Login Screen**.
  - Contrast polish on all badges, tables, inputs, stat cards, and action buttons.

Phase 9: Quality Assurance & Automated Testing

9.1 Objectives
Verify system integrity, deterministic seed state, cryptographic password hashing, role boundaries, and leave calculations across automated test runners.

9.2 Test Suite Execution (`tests/api.test.ts`)
17 automated integration assertions:

- ‘PASS‘: Initial database contains seeded users
- ‘PASS‘: Initial database contains seeded employees
- ‘PASS‘: Departments initialized correctly
- ‘PASS‘: HR user account exists
- ‘PASS‘: HR password hash verifies correctly
- ‘PASS‘: Invalid password is rejected
- ‘PASS‘: Sarah Jenkins has HR role
- ‘PASS‘: Marcus Vance has Manager role
- ‘PASS‘: Alex Rivera has Employee role
- ‘PASS‘: Employee Alex Rivera has recorded attendance history
- ‘PASS‘: Leave balance for Alex Rivera (casual) exists
- ‘PASS‘: Casual leave has 12 allocated days
- ‘PASS‘: Chloe Dupont leave record exists
- ‘PASS‘: Chloe leave is at Pending HR Approval stage after Manager approved it
- ‘PASS‘: Approval history logs Manager stage approval
- ‘PASS‘: Audit trail contains recorded events
- ‘PASS‘: Attendance correction audit log is present with metadata

Result: 17 Passed, 0 Failed (100% Passing).